

# Learning From $M$ -Tuple One-vs-All Confidence Comparison Data

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**Abstract**—Partial-label learning (PLL) studies classification with ambiguous supervision, where each instance is annotated by a candidate label set that contains the unknown true label. While PLL has been widely investigated, real-world annotation pipelines often provide richer structure than an unordered set: annotators or retrieval systems may implicitly rank candidates, yielding a *preferred label* that is more likely to be correct, and the mis-selection patterns can be class-dependent rather than uniform. Moreover, practical candidate sets may occasionally miss the true label due to human error or imperfect upstream models. Motivated by these observations, we propose a new weak-supervision paradigm, *preferred-label PLL in a one-vs-all view* (PLL-OVA), where each training instance is accompanied by a candidate set and a preferred label within the set. We model preferred-label generation through a flexible noisy-channel formulation that subsumes both *symmetric* and *asymmetric* mis-selection mechanisms, and we further incorporate the *missing-true-label* scenario to better reflect realistic annotation noise. Building on this formulation, we develop principled empirical risk minimization (ERM) procedures via risk rewriting, establish identifiability requirements under general channels, and derive practical learning objectives that remain statistically grounded. To improve optimization stability under negative-risk effects inherent to unbiased rewriting, we introduce *risk-correction* functions (e.g., rectified linear unit (ReLU)/absolute value correction function (ABS)-type corrections) that significantly enhance robustness and empirical performance. Extensive experiments on benchmark datasets demonstrate that PLL-OVA consistently improves upon standard PLL baselines, especially when candidate ambiguity is large or mis-selection is nonuniform, validating the effectiveness and practicality of the proposed framework.

**Index Terms**—One-vs-all, partial-label, risk correction, unbiased risk estimator (URE), weakly supervised learning.

## I. INTRODUCTION

**S**UPERVISED learning has achieved remarkable success across numerous domains but heavily relies on precisely labeled data, which is often costly to collect. In addition, due to certain unavoidable reasons (e.g., user privacy concerns, personal political orientation issues, sensitive topics related to religious beliefs, etc.), the labels of the samples cannot be directly obtained. As a result, classification methods using

weak supervision have been proposed [1]. A growing number of research has demonstrated that effective classifiers can be trained using weakly supervised information, including *positive-unlabeled (PU) learning* [2], [3], [4], [5], [6], [7], [8], which deals with situations where there is access to positive samples alongside a substantial pool of unlabeled data; *noisy-label learning* [9], [10], [11], [12], [13], [14], which represents a significant research direction in weakly supervised learning, leveraging inaccurate supervision; *positive-confidence (Pconf) learning* [15] train the classifier using the confidence score of the samples; *unlabeled-unlabeled (UU) learning* [16], [17], [18], [19] constructs learning model from entirely unlabeled data typically relies on weak distributional assumptions; *partial-label learning* (PLL) [20], [21], [22], [23], [24], [25] assumes that each instance could be associated with several potential labels, yet only one of these labels is the correct one; *complementary-label learning* [26], [27], [28], [29], [30], which is a typical learning framework that learning with “complementary labels,” where such labels signify that an instance is excluded from a certain class without explicitly stating its true class; *similarity-unlabeled (SU) learning* [31] and *similarity-confidence (Sconf) learning* [32] exploiting the similarity information between samples to construct a binary classifier; *pairwise confidence comparison (Pcomp) learning* [33], in which the classifier is learned by comparing the confidence scores between pairs of samples; *confidence-difference (ConfDiff) learning* [34] utilizes the difference in confidence scores as supervisory information.

One notable framework is *Pcomp learning* [33], which trains a binary classifier using pairwise comparison data instead of traditional pointwise labels. This framework assumes that for any pair of samples  $(x_a, x_b)$ , the confidence of instance  $x_a$  being a positive instance is higher than that of instance  $x_b$ . However, obtaining a large amount of pairwise data is a highly restrictive condition, which is difficult to meet adequately in practical applications. The samples in data tuples can appear not only as pairs but also as combinations of any number of samples. When dealing with these more realistic instance combinations, the *Pcomp* method fails to effectively utilize such weak supervision information to build a binary classifier. Therefore, this article conducts a more in-depth and comprehensive extension of the *Pcomp* method.

In pathological image analysis tasks, complex and rare diseases may present challenges in analyzing MRI images or tissue section images. On one hand, supervised methods require extensive pixel-level annotations, which are time-consuming and labor-intensive to generate. On the other hand, the highly heterogeneous morphology of medical imaging lesions makes the annotation process heavily reliant on the expertise and experience of pathologists. Variations in indi-

Received 6 January 2025; revised 24 June 2025, 27 September 2025, and 19 December 2025; accepted 9 March 2026. Date of publication 19 March 2026; date of current version 2 September 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62473327 and Grant 62373319, in part by the Natural Science Foundation of Hebei Province under Grant F2025203121 and Grant F2024203114, and in part by the Provincial Key Laboratory Performance Subsidy Project under Grant 22567612H. (Corresponding author: Junpeng Li.)

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This article has supplementary downloadable material available at <https://doi.org/10.1109/TNNLS.2026.3673280>, provided by the authors.

Digital Object Identifier 10.1109/TNNLS.2026.3673280

vidual expertise among experts often lead to unavoidable issues such as missed annotations or misannotations, reducing annotation accuracy. Only a limited number of highly knowledgeable and experienced pathologists can make rapid and accurate judgments for such diseases. Fortunately, less experienced or trainee physicians can leverage their knowledge to relatively easily identify an image within a group that is most probable to be classified as positive (i.e., the image that has a higher probability of signifying a disease compared to other images). Using these comparative relationships, the proposed one-vs-all confidence comparison learning method in this article can effectively address this problem.

This article begins with considering data in the form of tuples containing three instances and then generalizes it to tuples with an arbitrary number  $M$  of instances. The  $M$ -tuple data structure is represented as  $(x^1, x^2, \dots, x^M)$ , where there exists a comparative relationship within the group: the confidence that instance  $x^1$  is classified as a positive instance is greater than that of any other instance within the tuple. There are no restrictions on the size of  $M$ . We refer to this generalized learning framework as *M-tuple one-vs-all confidence comparison learning (MOVA)*.

Zhang et al. [35] reveal the vulnerability of label-noise learning under extreme noise and propose the *Robust DivideMix* framework as a stronger alternative. In comparison, the proposed *MOVA* framework focuses on tuple-level weak supervision and classifier construction, thereby advancing weakly supervised learning from a different perspective. Li et al. [36] present the *Msconf learning* framework, which models weak supervision at the  $M$ -tuple level via similarity confidence. By contrast, *MOVA learning* emphasizes “confidence comparison relations,” which are easier to obtain and provide finer-grained signals, leading to stronger discriminative power and wider applicability. Huang et al. [37] develop an unbiased risk estimator (URE) from pairwise data under the “non-all-negative” constraint combined with unlabeled samples. While effective in pairwise settings, it remains limited compared with the more general  $M$ -tuple modeling enabled by *MOVA*, which achieves stronger generalization through “one-vs-all” confidence comparisons.

The main contributions of this article are outlined below.

- 1) We propose a generalized learning framework termed as “*MOVA learning*,” featuring extensive practical uses.
- 2) We derive a URE  $\hat{R}_{MOVA}(g)$  based on the distribution of *MOVA* data tuples, establish its generalization error bound, and demonstrate that our learning method can achieve optimal parameter convergence rate.
- 3) The corrected risk was constructed using a correct function, and a theoretical proof of the consistency of the corrected risk estimator was provided.
- 4) Experiments based on *MOVA Learning* with triplets and quadruplets on a large number of datasets validate the rationality and effectiveness of the proposed methods.

The structure of this article is laid out in the following manner. Section II reviews the problem settings of traditional supervised learning and Pcomp learning. Section III provides a detailed explanation of the data generation process and the derivation of the risk function under the proposed *MOVA learning* framework. Section IV introduces a risk correction

method, which mitigates over-fitting during the training phase by enclosing the risk function within a correction function. Section V demonstrates the feasibility of the proposed method with experimental results from various benchmark datasets. Section VI presents the research conclusions of this work, while Section VII outlines the future research directions.

Proofs of the relevant lemmas and theorems mentioned in this article are provided in the Supplementary Material.

## II. PRELIMINARIES

In this section, a concise overview of the traditional binary classification problem setup is provided, along with a review of the background of Pcomp learning.

### A. Ordinary Binary Classification

In ordinary binary classification,  $\mathcal{X} \subset \mathbb{R}^d$  represents the feature space, and  $Y = \{-1, +1\}$  denotes the label space comprising two classes. Each instance  $(x, y)$  is sampled from the joint probability distribution characterized by the density  $p(x, y)$ . The primary objective is to derive a classifier  $g(\cdot) : \mathcal{X} \rightarrow \mathbb{R}$  capable of minimizing the classification risk

$$R(g) = \mathbb{E}_{p(x,y)} [\ell(g(x), y)] \quad (1)$$

where  $\mathbb{E}_{p(x,y)}$  denotes the expectation over  $p(x, y)$ , and  $\ell(\cdot, \cdot) : \mathbb{R} \times Y \rightarrow \mathbb{R}^+$  is a loss function that assesses the accuracy of the classifier in estimating the true class label. Let  $\pi_p = p(y = +1)$  [ $\pi_n = p(y = -1)$ ] denote the class-prior probability of the positive data (negative data). Then the equivalent expression of classification risk (1) can be expressed as

$$R(g) = \pi_p \mathbb{E}_p [\ell(g(x), +1)] + \pi_n \mathbb{E}_n [\ell(g(x), -1)] \quad (2)$$

where  $\mathbb{E}_p[\cdot]$  and  $\mathbb{E}_n[\cdot]$  are expectations over class-conditional probability density with  $p_p(x) = p(x|y = +1)$  and  $p_n(x) = p(x|y = -1)$ , respectively.

### B. Pcomp Learning

Data pairs with the weak supervision information suggest that instance  $x^a$  has a higher confidence of being classified as a positive instance compared to  $x^b$  [i.e.,  $p(y^a = +1|x^a) \geq p(y^b = +1|x^b)$ ] is defined as:  $\mathcal{D}_{Pcomp} = \{(x_i^a, x_i^b)\}_{i=1}^n$ , where  $(x_i^a, x_i^b)$  represents the  $i$ th group of unlabeled data pair, and  $n$  denotes the number of data pairs.

According to the weak supervision information within the *Pcomp learning*, *Pcomp* data pairs are assumed to be sampled from data pairs whose labels only have three forms, i.e.,  $\{(+1, -1), (+1, +1), (-1, -1)\}$ . Pointwise data in the data pairs are independently and identically drawn from the distribution  $\mathcal{D}_p = \{x_i^a\}_{i=1}^n$  and  $\mathcal{D}_n = \{x_i^b\}_{i=1}^n$ , respectively. During the data collection phase, labelers initially select two unlabeled data points independently of the marginal distribution  $p_p(x^a)$  and  $p_n(x^b)$ . Then, they assign a confidence comparison relationship to them.

**Theorem 1:** [33]: Pairwise example  $(x_i^a, x_i^b)$  in the dataset  $\mathcal{D}_{Pcomp}$  are independently and identically drawn from the distribution  $p_{Pcomp}(x^a, x^b)$ , which can be expressed as

$$p_{Pcomp}(x^a, x^b) = [\pi_p^2 p_p(x^a) p_p(x^b) + \pi_n^2 p_n(x^a) p_n(x^b) + \pi_p \pi_n p_p(x^a) p_n(x^b)] / (\pi_p^2 + \pi_n^2 + \pi_p \pi_n) \quad (3)$$

where

$$p_p(x^a) = p(x^a|y^a = +1), p_p(x^b) = p(x^b|y^b = +1) \\ p_n(x^a) = p(x^a|y^a = -1), p_n(x^b) = p(x^b|y^b = -1).$$

**Theorem 2 [33]:** The classification risk representation using the pcomp data is as follows:

$$R_{Pcomp}(g) = \mathbb{E}_{\hat{p}_p(x^a)} [\ell(g(x^a), +1) - \pi_p \ell(g(x^a), -1)] \\ + \mathbb{E}_{\hat{p}_n(x^b)} [-\pi_n \ell(g(x^b), +1) + \ell(g(x^b), -1)] \quad (4)$$

where

$$\hat{p}_p(x^a) = \frac{\pi_p}{\pi_p + \pi_n^2} p_p(x^a) + \frac{\pi_n^2}{\pi_p + \pi_n^2} p_n(x^a) \\ \hat{p}_n(x^b) = \frac{\pi_p^2}{\pi_p^2 + \pi_n} p_p(x^b) + \frac{\pi_n}{\pi_p^2 + \pi_n} p_n(x^b).$$

### III. LEARNING FROM ONE-VS-ALL $M$ -TUPLE CONFIDENCE COMPARISON DATA

This section first details the generation processes for both triple and  $M$ -tuple data with comparative confidence information. Then, a URE is derived to train a binary classifier by empirical risk minimization (ERM). Fig. 1 illustrates the generation process of MOVA data and the derivation of the URE.

#### A. Data Generation Process of Confidence Comparison Data

1) *One-vs-All Triple Confidence Comparison Data:* The labels of data tuples with the confidence comparison that the first instance has a higher confidence of being a positive instance compared to all other instances have five cases as follows:

$$\{(+1, -1, -1), (+1, +1, -1), (+1, -1, +1), \\ (+1, +1, +1), (-1, -1, -1)\}.$$

Assuming the dataset of data tuples that satisfy the aforementioned comparative relationships is denoted as  $D_3 = \{(x_i^1, x_i^2, x_i^3)\}_{i=1}^{n_3}$ , the dataset is independently generated from the distribution described in Lemma 1.

**Lemma 1:** Data tuples in  $D_3$  are independently drawn from

$$\tilde{p}(x^1, x^2, x^3) \\ = [\pi_p \pi_p^2 p_p(x^1) p_n(x^2) p_n(x^3) + \pi_p^2 \pi_n p_p(x^1) p_p(x^2) p_n(x^3) \\ + \pi_p^2 \pi_n p_p(x^1) p_n(x^2) p_p(x^3) + \pi_p^3 p_p(x^1) p_p(x^2) p_p(x^3) \\ + \pi_n^3 p_n(x^1) p_n(x^2) p_n(x^3)] / [\pi_p \pi_n^2 + 2\pi_p^2 \pi_n + \pi_p^3 + \pi_n^3]. \quad (5)$$

The pointwise samples  $\{x_i^1\}_{i=1}^{n_3}$  in  $\tilde{D}_1$ ,  $\{x_i^2\}_{i=1}^{n_3}$  in  $\tilde{D}_2$  and  $\{x_i^3\}_{i=1}^{n_3}$  in  $\tilde{D}_3$  are independently and identically generated from the marginal distributions  $\tilde{p}_1(x^1)$ ,  $\tilde{p}_2(x^2)$ , and  $\tilde{p}_3(x^3)$ , respectively. These probability distributions can be further decomposed into a weighted sum of positive and negative class components, providing a clearer representation of the underlying data generation process.

**Lemma 2:**  $\tilde{p}_1(x^1)$ ,  $\tilde{p}_2(x^2)$ ,  $\tilde{p}_3(x^3)$  can be expressed as

$$\tilde{p}_1(x^1) = \tilde{\pi}_{1p} p_p(x^1) + \tilde{\pi}_{1n} p_n(x^1)$$

$$\tilde{p}_2(x^2) = \tilde{\pi}_{2p} p_p(x^2) + \tilde{\pi}_{2n} p_n(x^2) \\ \tilde{p}_3(x^3) = \tilde{\pi}_{3p} p_p(x^3) + \tilde{\pi}_{3n} p_n(x^3)$$

where

$$\tilde{\pi}_{1p} = \frac{\pi_p \pi_n^2 + 2\pi_p^2 \pi_n + \pi_p^3}{\pi_p \pi_n^2 + 2\pi_p^2 \pi_n + \pi_p^3 + \pi_n^3} \\ \tilde{\pi}_{1n} = \frac{\pi_n^3}{\pi_p \pi_n^2 + 2\pi_p^2 \pi_n + \pi_p^3 + \pi_n^3} \\ \tilde{\pi}_{2p} = \frac{\pi_p^2 \pi_n + \pi_p^3}{\pi_p \pi_n^2 + 2\pi_p^2 \pi_n + \pi_p^3 + \pi_n^3} \\ \tilde{\pi}_{2n} = \frac{\pi_p \pi_n^2 + \pi_p^2 \pi_n + \pi_n^3}{\pi_p \pi_n^2 + 2\pi_p^2 \pi_n + \pi_p^3 + \pi_n^3} \\ \tilde{\pi}_{3p} = \frac{\pi_p^2 \pi_n + \pi_p^3}{\pi_p \pi_n^2 + 2\pi_p^2 \pi_n + \pi_p^3 + \pi_n^3} \\ \tilde{\pi}_{3n} = \frac{\pi_p \pi_n^2 + \pi_p^2 \pi_n + \pi_n^3}{\pi_p \pi_n^2 + 2\pi_p^2 \pi_n + \pi_p^3 + \pi_n^3}.$$

The comparative relationship where the first instance in a tuple has a higher confidence of being a positive instance compared to any other instance holds true for tuples of any size. Therefore, the size of such data tuples is not limited to three but extends to tuples of any size  $M$ .

Consequently, this article extends the concept of triplets to develop a generalized learning framework that constructs classifiers based on the confidence comparison relationships of  $M$  tuples. The data generation process involving confidence comparison relationships among  $M$  tuples is as follows:

2) *One-vs-All  $M$ -Tuple Confidence Comparison Data:* Extending the comparative relationships among the aforementioned three instances to comparative relationships among  $M$ -tuples, that is, instance  $x^1$  has a higher confidence of being a positive instance compared to the remaining instances  $x^2, \dots, x^M$ . This article denotes the dataset of data tuples that satisfy this comparative relationship as  $D_M = \{(x_i^1, x_i^2, \dots, x_i^M)\}_{i=1}^{n_M}$ . The distribution that  $D_M$  follows is shown in Lemma 3 below.

**Lemma 3:** Data tuples in  $D_M$  are independently drawn from

$$\hat{p}(x^1, x^2, \dots, x^M) \\ = \left[ \pi_p p_p(x^1) \sum_{t_1, t_2, \dots, t_{M-1} \in \{+1, -1\}} \prod_{i=2}^M [\mathbb{I}_{+}^{t_{i-1}} \pi_p p_p(x^i) + \mathbb{I}_{-}^{t_{i-1}} \pi_n p_n(x^i)] \right. \\ \left. + \pi_n^M \prod_{j=1}^M p_n(x^j) \right] / \left[ \sum_{k=0}^{M-1} (C_{M-1}^k \pi_p^{k+1} \pi_n^{M-1-k}) + \pi_n^M \right] \quad (6)$$

where  $\mathbb{I}_{+}^{t_{i-1}} = \mathbb{I}(t_{i-1} = +1)$ ,  $\mathbb{I}_{-}^{t_{i-1}} = \mathbb{I}(t_{i-1} = -1)$ .  $\mathbb{I}(\cdot)$  is an indicator function that takes the value of 1 when  $t_{i-1} = +1$  or  $t_{i-1} = -1$  holds true, and 0 otherwise.

The pointwise samples  $\{x_i^1\}_{i=1}^{n_M}$  in  $\hat{D}_1$ ,  $\{x_i^2\}_{i=1}^{n_M}$  in  $\hat{D}_2$ ,  $\dots$  and  $\{x_i^M\}_{i=1}^{n_M}$  in  $\hat{D}_M$  are independently and identically generated from the marginal distributions  $\hat{p}_1(x^1)$ ,  $\hat{p}_2(x^2)$ ,  $\dots$ , and  $\hat{p}_M(x^M)$ , respectively. These distributions serve as the probabilistic foundations for modeling the data, encapsulating both the class priors and the conditional probabilities associated with the samples. To further characterize the relationship

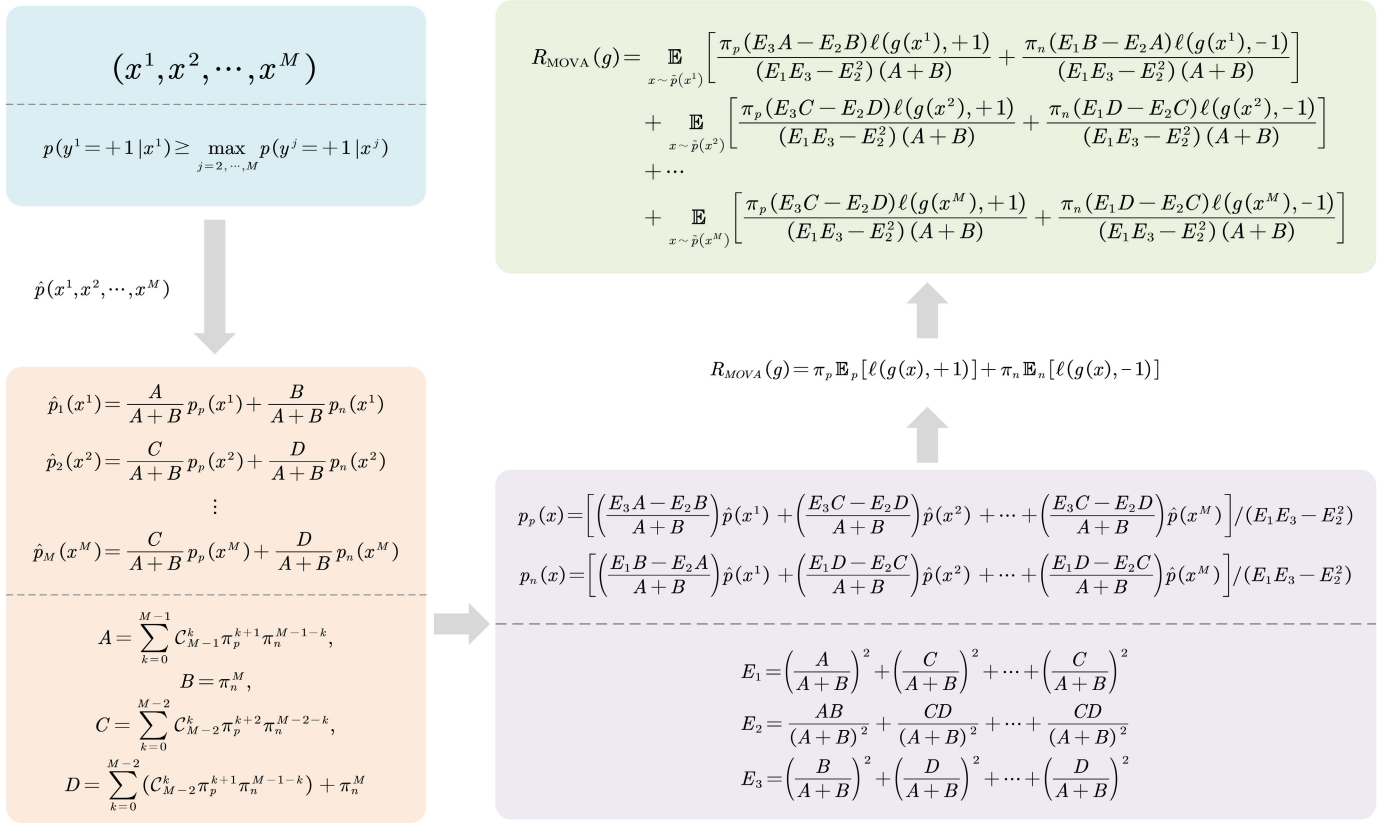


Fig. 1. Processes involved in generating MOVA data and deriving the URE.

between the distributions and their components,  $\hat{p}_1(x^1)$ ,  $\hat{p}_2(x^2)$ , ...,  $\hat{p}_M(x^M)$  can be represented as weighted combinations of the positive and negative class distributions. This decomposition enables a more detailed analysis of the data generation process.

**Lemma 4:**  $\hat{p}_1(x^1)$ ,  $\hat{p}_2(x^2)$ , ...,  $\hat{p}_M(x^M)$  can be expressed as

$$\begin{aligned} \hat{p}_1(x^1) &= \frac{A}{A+B}p_p(x^1) + \frac{B}{A+B}p_n(x^1) \\ \hat{p}_2(x^2) &= \frac{C}{A+B}p_p(x^2) + \frac{D}{A+B}p_n(x^2) \\ &\vdots \\ \hat{p}_M(x^M) &= \frac{C}{A+B}p_p(x^M) + \frac{D}{A+B}p_n(x^M) \end{aligned}$$

where

$$\begin{aligned} A &= \sum_{k=0}^{M-1} C_{M-1}^k \pi_p^{k+1} \pi_n^{M-1-k}, & B &= \pi_n^M \\ C &= \sum_{k=0}^{M-2} C_{M-2}^k \pi_p^{k+2} \pi_n^{M-2-k} \\ D &= \sum_{k=0}^{M-2} (C_{M-2}^k \pi_p^{k+1} \pi_n^{M-1-k}) + \pi_n^M. \end{aligned}$$

The specific expressions for the distribution of positive samples  $p_p(x)$  and the distribution of negative samples  $p_n(x)$  are provided by Lemma 5 below.

**Lemma 5:** The distribution of positive samples, denoted as  $p_p(x)$ , and the distribution of negative samples, denoted as

$p_n(x)$ , can be expressed, respectively, by  $\hat{p}(x^1)$ ,  $\hat{p}(x^2)$ , ..., and  $\hat{p}(x^M)$

$$\begin{aligned} p_p(x) &= \left[ \left( \frac{E_3A - E_2B}{A+B} \right) \hat{p}(x^1) + \left( \frac{E_3C - E_2D}{A+B} \right) \hat{p}(x^2) \right. \\ &\quad \left. + \dots + \left( \frac{E_3C - E_2D}{A+B} \right) \hat{p}(x^M) \right] / (E_1E_3 - E_2^2) \\ p_n(x) &= \left[ \left( \frac{E_1B - E_2A}{A+B} \right) \hat{p}(x^1) + \left( \frac{E_1D - E_2C}{A+B} \right) \hat{p}(x^2) \right. \\ &\quad \left. + \dots + \left( \frac{E_1D - E_2C}{A+B} \right) \hat{p}(x^M) \right] / (E_1E_3 - E_2^2) \end{aligned}$$

where

$$\begin{aligned} E_1 &= \left( \frac{A}{A+B} \right)^2 + \left( \frac{C}{A+B} \right)^2 + \dots + \left( \frac{C}{A+B} \right)^2 \\ E_2 &= \frac{AB}{(A+B)^2} + \frac{CD}{(A+B)^2} + \dots + \frac{CD}{(A+B)^2} \\ E_3 &= \left( \frac{B}{A+B} \right)^2 + \left( \frac{D}{A+B} \right)^2 + \dots + \left( \frac{D}{A+B} \right)^2. \end{aligned}$$

## B. URE With MOVA Data

**Theorem 3:** The classification risk  $R(g)$  (1) is presented using MOVA data

$$\begin{aligned} R_{\text{MOVA}}(g) &= \pi_p \mathbb{E}_p[\ell(g(x), +1)] + \pi_n \mathbb{E}_n[\ell(g(x), -1)] \\ &= \mathbb{E}_{x \sim \hat{p}(x^1)} \left[ \frac{\pi_p(E_3A - E_2B)\ell(g(x^1), +1)}{(E_1E_3 - E_2^2)(A+B)} \right] \end{aligned}$$

$$\begin{aligned}
& + \frac{\pi_n(E_1B - E_2A)\ell(g(x^1), -1)}{(E_1E_3 - E_2^2)(A+B)} \Bigg] \\
& + \mathbb{E}_{x \sim \tilde{p}(x^2)} \left[ \frac{\pi_p(E_3C - E_2D)\ell(g(x^2), +1)}{(E_1E_3 - E_2^2)(A+B)} \right. \\
& + \frac{\pi_n(E_1D - E_2C)\ell(g(x^2), -1)}{(E_1E_3 - E_2^2)(A+B)} \Bigg] + \dots \\
& + \mathbb{E}_{x \sim \tilde{p}(x^M)} \left[ \frac{\pi_p(E_3C - E_2D)\ell(g(x^M), +1)}{(E_1E_3 - E_2^2)(A+B)} \right. \\
& + \frac{\pi_n(E_1D - E_2C)\ell(g(x^M), -1)}{(E_1E_3 - E_2^2)(A+B)} \Bigg]. \quad (7)
\end{aligned}$$

Building on the conclusions derived from Theorem 3, we can formally establish the expression for the empirical risk associated with *MOVA* data. This result follows naturally from the theoretical framework outlined earlier, as the comparative relationships among data tuples provide a direct basis for the empirical risk formulation:

$$\begin{aligned}
& \hat{R}_{MOVA}(g) \\
& = \frac{1}{n} \sum_{i=1}^n \left[ \pi_p \left( \frac{E_3A - E_2B}{A+B} \right) \ell(g(x_i^1), +1) \right. \\
& + \pi_p \left( \frac{E_3C - E_2D}{A+B} \right) \ell(g(x_i^2), +1) + \dots \\
& + \pi_p \left( \frac{E_3C - E_2D}{A+B} \right) \ell(g(x_i^M), +1) \\
& + \pi_n \left( \frac{E_1B - E_2A}{A+B} \right) \ell(g(x_i^1), -1) \\
& + \pi_n \left( \frac{E_1D - E_2C}{A+B} \right) \ell(g(x_i^2), -1) + \dots \\
& + \left. \pi_n \left( \frac{E_1D - E_2C}{A+B} \right) \ell(g(x_i^M), -1) \right] / (E_1E_3 - E_2^2). \quad (8)
\end{aligned}$$

### C. Estimation Error Bound

An estimation error bound is established for *MOVA learning* to elucidate the convergence of the proposed risk estimator  $\hat{R}_{MOVA}(g)$  as defined in (8). Given that the algorithm design separates true risk estimation from ERM and decouples estimation error from model-independent complexity measures and their associated constraints across models, this article employs Rademacher complexity as a standard measure of model complexity.

**Definition 1: (Rademacher Complexity)** [38]: Let  $Z_1, \dots, Z_n$  be i.i.d. random variables drawn from a probability distribution with density  $\mu$ ,  $\mathcal{G} = \{g : \mathcal{X} \rightarrow \mathbb{R}\}$  be a class of measurable functions. Then, the Rademacher complexity of  $\mathcal{G}$  is defined as

$$\mathfrak{R}(\mathcal{G}) = \mathbb{E}_{Z_1, \dots, Z_n \sim \mu} \mathbb{E}_{\sigma} \left[ \sup_{g \in \mathcal{G}} \frac{1}{n} \sum_{i=1}^n \sigma_i g(Z_i) \right] \quad (9)$$

where  $n$  is a positive integer, and  $\sigma = (\sigma_1, \dots, \sigma_n)$  are Rademacher variables that taking from  $\{-1, +1\}$  uniformly.

To derive an estimation error bound, establishing a generalization error bound is crucial. The proof of the generalization error bound hinges on Talagrand's lemma [38], which utilizes

the empirical Rademacher complexity of a hypothesis set to bound the empirical Rademacher complexity of the composition of that model set  $\mathcal{G}$  with a Lipschitz-continuous loss function  $\ell$ .

Assume that there exists a constant  $C_f > 0$  that  $\sup_{g \in \mathcal{G}} \|g\|_{\infty} \leq C_f$  and  $C_{\ell} > 0$  such that  $\sup_{|z| \leq C_f} \ell(z, y) \leq C_{\ell}$  holds for all  $y$ . It is also assumed that the loss function  $\ell(z, y)$  is Lipschitz continuous for  $z$  with  $|z| \leq C_f$  and  $y$  with a Lipschitz constant  $L_{\ell}$ . Let  $g^* = \arg \min_{g \in \mathcal{G}} R(g)$  be the minimizer of true risk in (1), and  $\hat{g} = \arg \min_{g \in \mathcal{G}} \hat{R}(g)$  be the minimizer of empirical risk in (8).

**Lemma 6:** Let  $C_{\mathcal{G}}$  represents a non-negative constant. It is assumed that the following inequality holds for the set of models  $\mathcal{G}$  and any given probability density  $\mu$  defined over the data space:

$$\mathfrak{R}(\mathcal{G}) \leq \frac{C_{\mathcal{G}}}{\sqrt{n}}. \quad (10)$$

Subsequently, the estimation error bound for *MOVA learning* is established by leveraging Rademacher complexity of *MOVA* data tuples, Talagrand's lemma, and McDiarmid's inequality [38].

**Theorem 4:** For any  $\delta > 0$ , the estimation error bound holds with probability at least  $1 - \delta$

$$\begin{aligned}
& R(\hat{g}_{MOVA}) - R(g^*) \\
& \leq \left( \left| \frac{\pi_p(E_3A - E_2B)}{(E_1E_3 - E_2^2)(A+B)} \right| + \left| \frac{\pi_n(E_1B - E_2A)}{(E_1E_3 - E_2^2)(A+B)} \right| \right. \\
& + \left| \frac{\pi_p(E_3C - E_2D)}{(E_1E_3 - E_2^2)(A+B)} \right| + \left| \frac{\pi_n(E_1D - E_2C)}{(E_1E_3 - E_2^2)(A+B)} \right| \\
& + \dots + \left| \frac{\pi_p(E_3C - E_2D)}{(E_1E_3 - E_2^2)(A+B)} \right| \\
& + \left. \left| \frac{\pi_n(E_1D - E_2C)}{(E_1E_3 - E_2^2)(A+B)} \right| \right) \\
& * \left( \frac{4\rho C_{\mathcal{G}}}{\sqrt{n}} + 2C_{\ell} \sqrt{\frac{\log \frac{2}{\delta}}{2n}} \right). \quad (11)
\end{aligned}$$

Theorem 4 ensures the consistency of our proposed method, expressed as  $R(\hat{g}_{MOVA}) \rightarrow R(g^*)$  as  $n \rightarrow \infty$ . The estimation error bound for *MOVA learning* converges at the rate of  $\mathcal{O}_p(1/\sqrt{n})$  ( $\mathcal{O}(\cdot)$  denotes the order in probability), demonstrating the optimal parametric rate for ERM under minimal assumptions [39].

## IV. RISK CORRECTION METHOD

In Section III, we derive a URE for *MOVA* data via risk rewriting. Consistency analysis ensures that, with sufficient samples, the empirical risk approximates the true risk with high probability. However, in practice, the estimator is prone to overfitting, especially with complex models, and this issue cannot be resolved by standard regularization. Our experiments show that overfitting consistently coincides with negative training risk (Figs. 2–5), a phenomenon also noted in [17]. To address this, a correction function  $g(x)$  is applied to enforce non-negativity of the empirical risk.

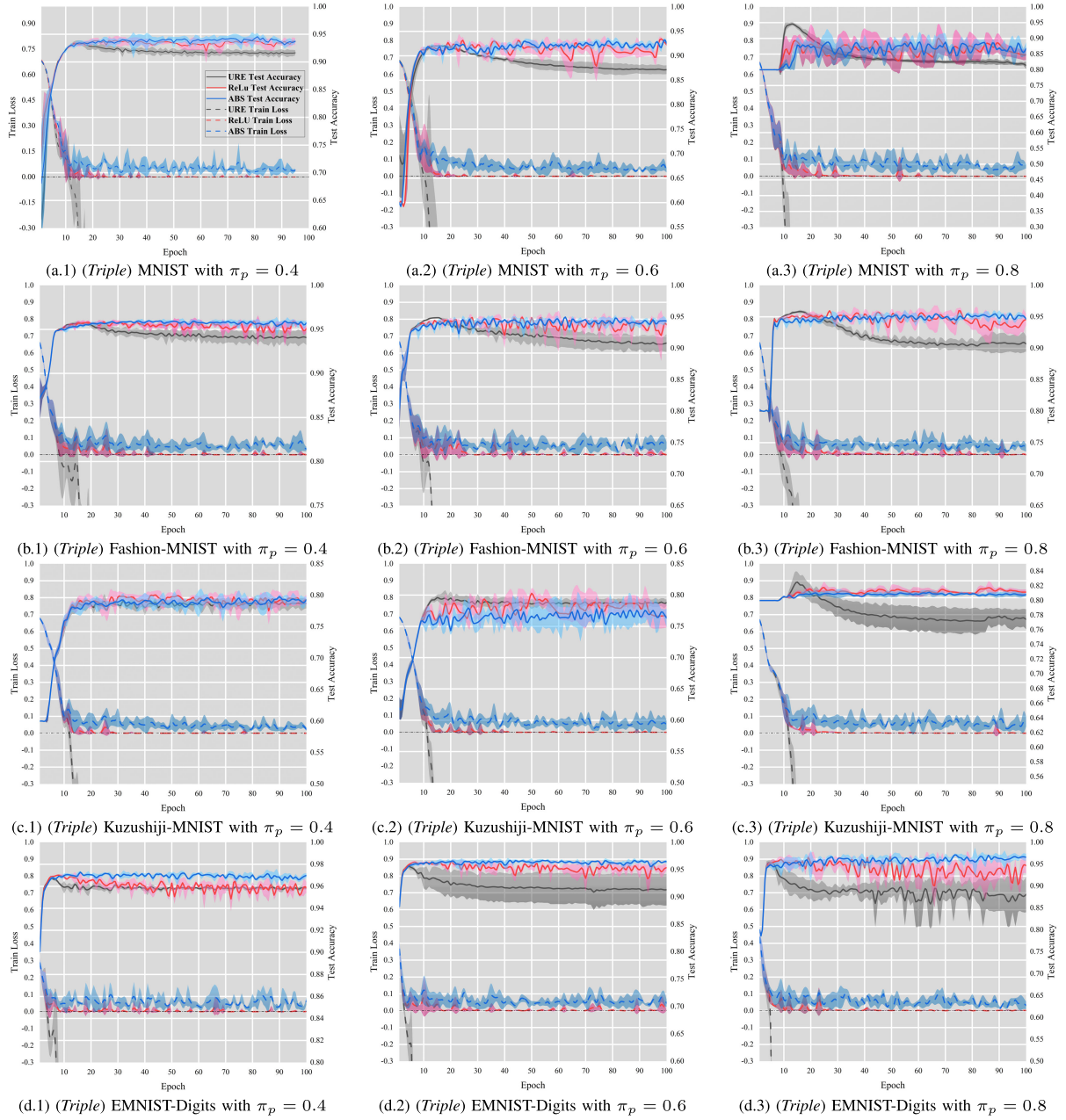


Fig. 2. Classification performance of *Triple-MOVA learning* with logistic loss is evaluated on four low-dimensional datasets (e.g., MNIST, Fashion-MNIST, Kuzushiji-MNIST, and EMNIST-Digits) with  $28 \times 28$  feature dimension. The mean accuracy is represented by dark colors, while the standard deviation is indicated by light colors. (a.1)–(a.3) Outcomes of *Triple-MOVA learning* on MNIST. (b.1)–(b.3) Outcomes of *Triple-MOVA learning* on Fashion-MNIST. (c.1)–(c.3) Results of *Triple-MOVA learning* on Kuzushiji-MNIST. (d.1)–(d.3) Results of *Triple-MOVA learning* on EMNIST-Digits.

The use of a simple  $\max(\cdot, 0)$  operator discards training samples that contribute to negative risk, potentially eliminating valuable information that should not be overlooked. Therefore, this article introduces a consistent correction function to ensure the asymptotic consistency of the risk estimator. A consistent correction function must satisfy three conditions: 1) Lipschitz continuity; 2) non-negativity of function values; and 3)  $f(x) = x$  for all  $x \leq 0$ . Lipschitz continuity bounds the function's variation rate, preventing abrupt output changes caused by noisy input fluctuations. When analyzing the deviation between empirical and true risk via McDiarmid's inequality, Lipschitz continuity is critical for deriving error bounds, ensuring the corrected risk estimator converges exponentially fast to the

true risk. Non-negativity directly avoids negative empirical risk values by enforcing positivity. The condition  $f(x) = x$  for  $x \leq 0$  retains the positive risk components, guaranteeing the asymptotic consistency of the corrected risk estimator.

Both the rectified linear unit (ReLU) and absolute value correction function (ABS) satisfy the above three consistency conditions and achieve the goal of eliminating negative risk. Therefore, this article selects the ReLU and ABS to encapsulate the empirical risk.

The expressions for these two correction functions are as follows:

$$\tilde{R}(g) = f(\hat{R}_p(g)) + f(\hat{R}_n(g)) \quad (12)$$

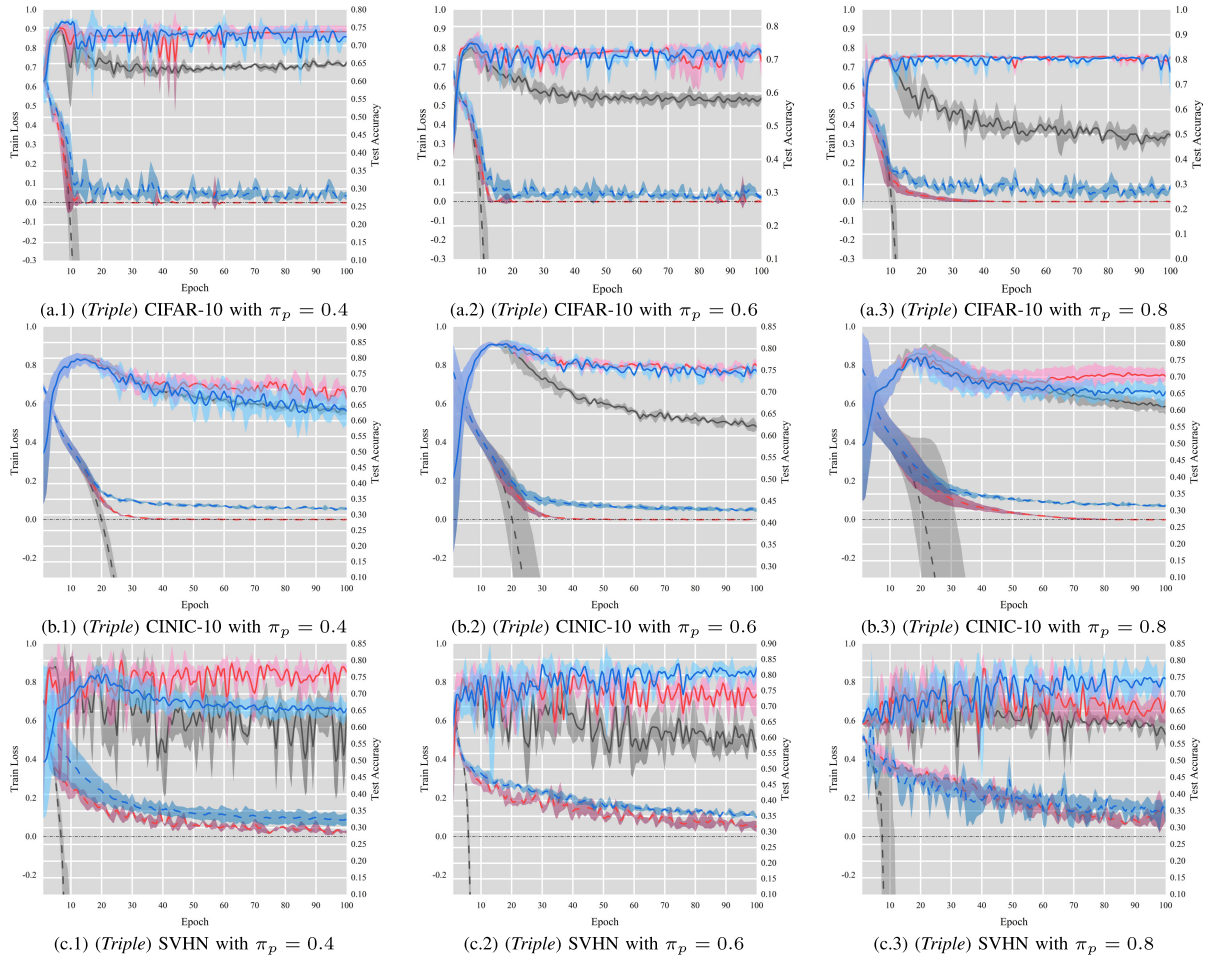


Fig. 3. Classification performance of *Triple-MOVA* learning with logistic loss is evaluated on three high-dimensional datasets (e.g., CIFAR-10, CINIC-10, and SVHN) with  $32 \times 32 \times 3$  feature dimension. The mean accuracy is represented by dark colors, while the standard deviation is indicated by light colors. (a.1)–(a.3) Outcomes of *Triple-MOVA* learning on CIFAR-10. (b.1)–(b.3) Outcomes of *Triple-MOVA* learning on CINIC-10. (c.1)–(c.3) Results of *Triple-MOVA* learning on SVHN.

where  $f(x) = \begin{cases} x, & x \geq 0, \\ k|x|, & x < 0. \end{cases}$  and  $k > 0$ . When  $k = 1$ , it corresponds to the absolute value correction function; when  $k = 0$ , it corresponds to the ReLU.

The representation of the corrected empirical risk is given by

$$\tilde{R}_{ReLU}(g) = \max\{0, \hat{R}_p(g)\} + \max\{0, \hat{R}_n(g)\} \quad (13)$$

$$\tilde{R}_{ABS}(g) = |\hat{R}_p(g)| + |\hat{R}_n(g)|. \quad (14)$$

The ReLU function truncates negative risk to zero, preserving only positive risk information. This simplifies computation and enhances model sparsity. The ABS function ensures non-negative risk values by taking absolute values, retaining the absolute information of risk. By enforcing non-negativity through correction functions, the model avoids overfitting to noise or outliers, thereby addressing overfitting issues.

Then, we give some theoretical analysis to show the consistency of the corrected risk estimator.

**Theorem 5:** [Consistency of  $\tilde{R}_{MOVA}(g)$ ] Suppose that there exists two positive constant  $\zeta$  and  $\eta$  satisfy that  $R_p(g) \geq \zeta$  and  $R_n(g) \geq \eta$ . As stated in Theorem 4, the bias of  $\tilde{R}_{MOVA}(g)$  decreases at an exponential rate as  $n \rightarrow \infty$

$$\mathbb{E}[\tilde{R}_{MOVA}(g)] - R(g) \leq (L_f + 1)(|p_1| + \dots + |p_M| + |n_1| + \dots + |n_M|) C_\ell \Delta. \quad (15)$$

In addition, with probability at least  $1 - \delta$

$$\begin{aligned} |\tilde{R}_{MOVA}(g) - R(g)| &\leq C_\ell L_f (|p_1| + \dots + |p_M| + |n_1| + \dots + |n_M|) \sqrt{\frac{\ln 2/\delta}{2n}} \\ &\quad + (L_f + 1)(|p_1| + \dots + |p_M| + |n_1| + \dots + |n_M|) C_\ell \Delta \end{aligned} \quad (16)$$

where

$$p_1 = \frac{\pi_p(E_3A - E_2B)}{(E_1E_3 - E_2^2)(A + B)}$$

$$p_2 = \dots = p_M = \frac{\pi_p(E_3C - E_2D)}{(E_1E_3 - E_2^2)(A + B)}$$

$$n_1 = \frac{\pi_n(E_1B - E_2A)}{(E_1E_3 - E_2^2)(A + B)}$$

$$n_2 = \dots = n_M = \frac{\pi_n(E_1D - E_2C)}{(E_1E_3 - E_2^2)(A + B)}$$

$$\Delta = \exp(-2n\zeta^2/C_\ell^2(|p_1|^2 + |p_2|^2 + \dots + |p_M|^2))$$

$$+ \exp(-2n\eta^2/C_\ell^2(|n_1|^2 + |n_2|^2 + \dots + |n_M|^2)).$$

Theorem 5 demonstrates that the convergence rate of  $\tilde{R}_{MOVA}(g) \rightarrow R(g)$  is equivalent to that of  $R(\hat{g}_{MOVA}) \rightarrow R(g^*)$ , i.e., the empirical risk minimizer  $\tilde{g}$  obtained via ERM achieves convergence to  $g^*$  with a rate of  $\mathcal{O}_p(1/\sqrt{n})$ , and Theorem 6 establishes the bound on its estimation error.

**Theorem 6 (Estimation Error Bound of  $\tilde{g}_{MOVA}$ ):** Based on the previously stated assumptions and notations, the following inequality is satisfied with a probability of at least  $1 - \delta$ :

$$\begin{aligned} & R(\tilde{g}_{MOVA}) - R(g^*) \\ & \leq 2C_\ell L_f (|p_1| + \dots + |p_M| + |n_1| + \dots + |n_M|) \sqrt{\frac{\ln 2/\delta}{2n}} \\ & \quad + 2(L_f + 1)(|p_1| + \dots + |p_M| + |n_1| + \dots + |n_M|) C_\ell \mathcal{A} \\ & \quad + \left( \left| \frac{\pi_p(E_3A - E_2B)}{(E_1E_3 - E_2^2)(A+B)} \right| + \left| \frac{\pi_n(E_1B - E_2A)}{(E_1E_3 - E_2^2)(A+B)} \right| \right. \\ & \quad + \left| \frac{\pi_p(E_3C - E_2D)}{(E_1E_3 - E_2^2)(A+B)} \right| + \left| \frac{\pi_n(E_1D - E_2C)}{(E_1E_3 - E_2^2)(A+B)} \right| \\ & \quad + \dots + \left| \frac{\pi_p(E_3C - E_2D)}{(E_1E_3 - E_2^2)(A+B)} \right| \\ & \quad + \left. \left| \frac{\pi_n(E_1D - E_2C)}{(E_1E_3 - E_2^2)(A+B)} \right| \right) \\ & \quad * \left( \frac{4\rho C_\mathcal{G}}{\sqrt{n}} + 2C_\ell \sqrt{\frac{\log \frac{2}{\delta}}{2n}} \right). \end{aligned} \quad (17)$$

Through the analysis of Theorem 6, it can be concluded that as the number of training samples  $n$  approaches infinity, the corrected risk converges to the true risk at an optimal rate of  $\mathcal{O}_p(1/\sqrt{n})$ .

## V. EXPERIMENTS

In this section, we first elaborate on three key aspects: the dataset used, the experimental parameter settings, and the experimental methods employed in this article, including the baseline methods. Finally, we present a detailed analysis of the experimental results and the training images generated from the experiments.

### A. Datasets

This article selects seven datasets for experiments to validate the authenticity and effectiveness of the proposed method. Since samples in each dataset may belong to multiple different classes, these datasets are commonly used for multiclassification problems. However, to align with the binary classification goal of this study, we manually reclassify the samples. The subsequent content describes the binary classification settings for each of the seven datasets.

**MNIST [40]:** It is a grayscale handwritten digits recognition dataset with an original feature dimension of  $28 \times 28$ . It consists of 60 000 training examples and 10 000 test examples. The label space encompasses digits 0–9. In our experimental setting, digits in even-numbered positions {0, 2, 4, 6, 8} are considered the positive class, while digits in odd-numbered positions {1, 3, 5, 7, 9} are considered the negative class.

**Fashion-MNIST [41]:** This dataset features grayscale images of fashion items, with each image having an original feature dimension of  $28 \times 28$ . It includes 60 000 training samples and 10 000 testing samples. The label space comprises the categories: {"T-shirt," "Pullover," "Dress," "Shirt," "Trouser," "Coat," "Sandal," "Sneaker," "Bag," "Ankle boot"}. In our experimental framework, the classes {"T-shirt," "Pullover," "Shirt," "Bag," "Coat"} are designated as the positive class, while {"Dress," "Trouser," "Sandal," "Sneaker," "Ankle boot"} are classified as the negative class.

**Kuzushiji-MNIST [42]:** This dataset consists of grayscale images of Japanese characters, each with an original feature dimension of  $28 \times 28$ . It contains 60 000 training samples and 10 000 testing samples. The label space includes the characters: {"Na," "Ha," "Ma," "Ya," "O," "Ki," "Su," "Tsu," "Re," "Wo"}. In our experimental setup, the characters {"O," "Re," "Na," "Ma," "Su"} are classified as the positive class, while {"Ha," "Ya," "Ki," "Tsu," "Wo"} are considered the negative class.

**EMNIST-Digits [43]:** The EMNIST-Digits dataset is a subset of the EMNIST dataset, featuring grayscale images of handwritten digits with an original feature dimension of  $28 \times 28$ . It comprises 240 000 training examples and 40 000 test examples. The label space includes digits 0–9. In our experimental setup, digits in even-numbered positions {0, 2, 4, 6, 8} are classified as the positive class, while digits in odd-numbered positions {1, 3, 5, 7, 9} are designated as the negative class.

**CIFAR-10 [44]:** This dataset features colorful object recognition images with an original feature dimension of  $32 \times 32 \times 3$ . It consists of 60 000 training examples and 10 000 test examples. The label space includes the categories: {"bird," "dog," "cat," "deer," "frog," "horse," "airplane," "automobile," "ship," "truck"}. In our experimental framework, {"bird," "dog," "cat," "deer," "frog," "horse"} are classified as the positive class, while {"airplane," "automobile," "ship," "truck"} are considered the negative class.

**CINIC-10 [45]:** The CINIC-10 dataset is constructed by combining CIFAR-10 with images downsampled from the ImageNet database to match the CIFAR-10 resolution of  $32 \times 32 \times 3$ . This results in higher data diversity and more complex visual features. It contains 180 000 training images and 90 000 test images. The label space includes the categories: {"bird," "dog," "cat," "deer," "frog," "horse," "airplane," "automobile," "ship," "truck"}. In our experimental framework, {"bird," "dog," "cat," "deer," "frog," "horse"} are classified as the positive class, while {"airplane," "automobile," "ship," "truck"} are considered the negative class.

**SVHN [46]:** The SVHN dataset is a classical benchmark for object recognition in computer vision. It contains  $32 \times 32 \times 3$  color images of house numbers extracted from Google Street View images. These images include real-world challenges such as varying lighting conditions, occlusions, and multiple house numbers per image, testing the model's ability to focus on local details. The dataset consists of 73 257 training images and 20 632 test images. The label space encompasses digits 0–9. In our experimental setting, digits in even-numbered positions {2, 3, 4, 5, 6, 7} are considered the positive class, while

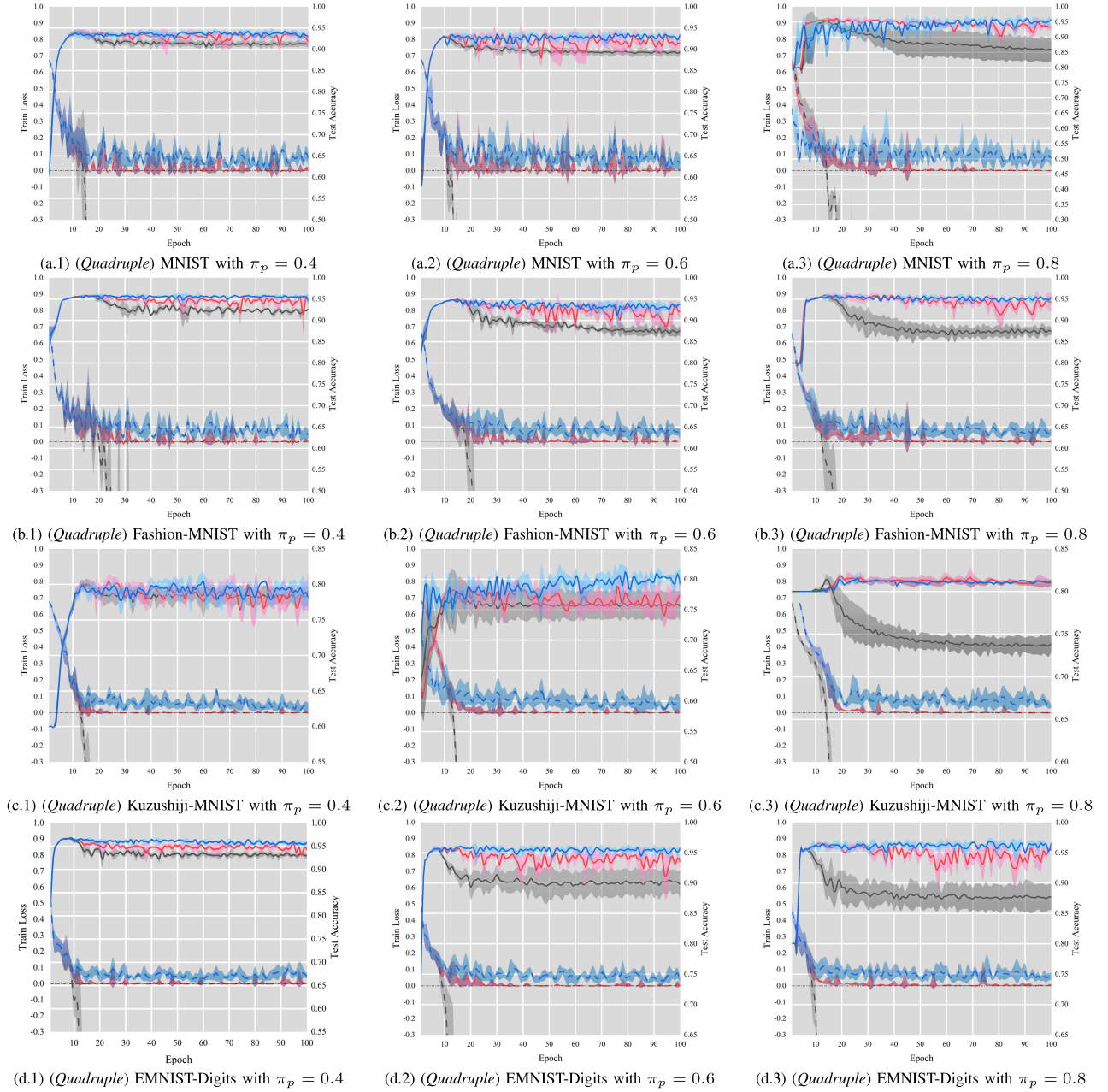


Fig. 4. Classification performance of *Quadruple-MOVA learning* with logistic loss is evaluated on four low-dimensional datasets (e.g., MNIST, Fashion-MNIST, Kuzushiji-MNIST, and EMNIST-Digits) with  $28 \times 28$  feature dimension. The mean accuracy is represented by dark colors, while the standard deviation is indicated by light colors. (a.1)–(a.3) Outcomes of *Quadruple-MOVA learning* on MNIST. (b.1)–(b.3) Outcomes of *Quadruple-MOVA learning* on Fashion-MNIST. (c.1)–(c.3) Results of *Quadruple-MOVA learning* on Kuzushiji-MNIST. (d.1)–(d.3) Results of *Quadruple-MOVA learning* on EMNIST-Digits.

digits in odd-numbered positions  $\{0, 1, 8, 9\}$  are considered the negative class.

### B. Experimental Setup

In this article, the learning rate was selected from  $\{5e^{-4}, 1e^{-5}, 1e^{-4}, 1e^{-3}\}$ , and the weight decay was set to  $\{1e^{-3}, 1e^{-2}\}$ . To clearly illustrate the training process, the number of training epochs was set to 100. For the four benchmark datasets—MNIST, Fashion-MNIST, Kuzushiji-MNIST, and EMNIST-Digits—a multilayer perceptron (MLP) with three hidden layers was used as the training model. In validating the proposed method on high-dimensional color image datasets comprising CIFAR-10, CINIC-10, and SVHN, ResNet-34 was

adopted as the training model. To evaluate the generality of the proposed method, comparative experiments were conducted under three class prior conditions, specifically  $\{0.4, 0.6, 0.8\}$ .

All methods were implemented in Pytorch, and the experiments were conducted on an NVIDIA GeForce RTX 4060.

### C. Methods

The experimental method in this study follows the principle of ERM by minimizing the proposed empirical risk function (7) and two risk-corrected functions obtained through the ReLU and ABS correction functions [see (13) and (14)]. In the experimental analysis, the three methods are abbreviated as URE, ReLU, and ABS.

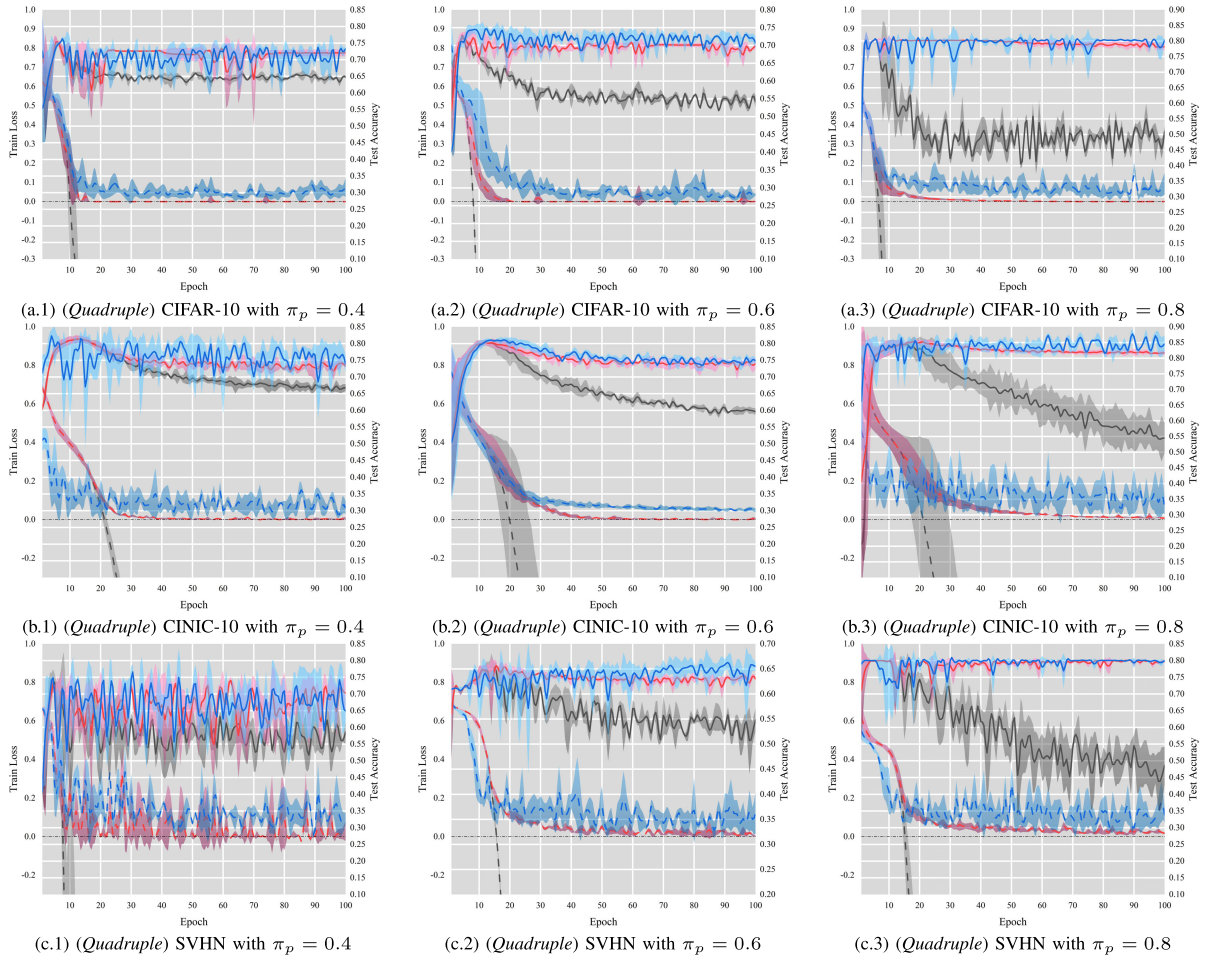


Fig. 5. Classification performance of *Quadruple-MOVA* learning with logistic loss is evaluated on three high-dimensional datasets (e.g., CIFAR-10, CINIC-10, and SVHN) with  $32 \times 32 \times 3$  feature dimension. The mean accuracy is represented by dark colors, while the standard deviation is indicated by light colors. (a.1)–(a.3) Outcomes of *Triple-MOVA* learning on CIFAR-10. (b.1)–(b.3) Outcomes of *Triple-MOVA* learning on CINIC-10. (c.1)–(c.3) Results of *Triple-MOVA* learning on SVHN.

#### D. Baseline Methods

**WCL:** We adopt the weakly supervised variant of *SimCLR* [47], namely, *WCL* [48], as a baseline. Similar to *SimCLR*, *WCL* follows a two-stage paradigm of self-supervised pretraining with the NT-Xent loss and subsequent linear probing, while exploiting weak supervision signals to guide the construction of positive and negative pairs. In our implementation, *WCL* is further adapted with a one-vs-all construction and a winner-only anchor strategy to align with the assumptions of the proposed *MOVA* framework. For grayscale datasets, we adopt a lightweight MLP model, whereas for color datasets, a ResNet-18 backbone is used.

**Siamese:** The Siamese network is a specialized neural network architecture commonly used to compare the similarity between two input samples. It learns feature representations of the inputs by sharing weights. In this study,  $M$  is set to 3 and 4 to compare pairwise similarities among three and four instances, respectively.

**Contrastive:** Contrastive loss is a loss function used to train Siamese networks by reducing the distance between feature vectors of similar samples while increasing the distance between those of dissimilar samples.

Since Siamese networks and contrastive loss can only utilize feature vectors rather than class labels, in the contrast experiments of “Siamese” and “Contrastive,”  $M$  prototypes with different labels are randomly selected. The classification accuracy is then calculated by comparing the similarity between the classified instance and these  $M$  prototypes.

**K-Means:** Clustering is a type of unsupervised learning, where data is trained without instance labels. K-Means clustering, one of the most classic clustering algorithms, divides samples into  $K$  clusters through an iterative process. Since this study focuses on a binary classification problem, the value of  $K$  is set to 2 in the experiments. The training data in this study consists of combinations of multiple samples (triplets and quadruplets). To ensure that the training instance vectors retain the overall information of the three images, the feature vectors of the three images are concatenated into a longer feature vector. This new feature vector is then used as a single input instance for the K-Means clustering algorithm.

#### E. Experimental Results

The experimental results, as shown in Tables I and II, clearly indicate that the proposed ReLU and ABS methods

TABLE I  
CLASSIFICATION ACCURACY (MEAN $\pm$ STD) OF THREE MOVA LEARNING METHODS ( $M = 3$ ) IS REPORTED FOR FIVE DATASETS, AVERAGED OVER 100 EPOCHS AND THREE TRIALS, UNDER THREE VARYING CLASS-PRIORS

| $\pi_p$ | Methods     | MNIST                             | Fashion-MNIST                     | Kuzushiji-MNIST                   | EMNIST-Digits                     | CIFAR-10                          | CINIC-10                          | SVHN                              |
|---------|-------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
| 0.4     | URE         | 91.93 $\pm$ 0.50%                 | 94.50 $\pm$ 0.54%                 | 78.56 $\pm$ 0.70%                 | 95.84 $\pm$ 0.12%                 | 64.37 $\pm$ 0.91%                 | 68.85 $\pm$ 1.63%                 | 61.11 $\pm$ 1.11%                 |
|         | ReLU        | 93.40 $\pm$ 0.40%                 | 95.37 $\pm$ 0.23%                 | <b>79.11<math>\pm</math>0.64%</b> | 95.97 $\pm$ 0.21%                 | <b>72.91<math>\pm</math>1.55%</b> | <b>71.59<math>\pm</math>2.82%</b> | 73.86 $\pm$ 0.44%                 |
|         | ABS         | <b>93.77<math>\pm</math>0.55%</b> | <b>95.72<math>\pm</math>0.21%</b> | 78.80 $\pm$ 0.39%                 | <b>96.94<math>\pm</math>0.09%</b> | 72.53 $\pm$ 0.88%                 | 68.94 $\pm$ 3.88%                 | <b>75.51<math>\pm</math>1.69%</b> |
|         | WCL         | 91.11 $\pm$ 0.22%                 | 93.54 $\pm$ 0.86%                 | 72.87 $\pm$ 1.07%                 | 92.09 $\pm$ 0.39%                 | 71.63 $\pm$ 1.03%                 | 71.16 $\pm$ 0.94%                 | 74.57 $\pm$ 1.69%                 |
|         | Siamese     | 52.41 $\pm$ 1.48%                 | 51.93 $\pm$ 1.33%                 | 60.33 $\pm$ 0.51%                 | 61.21 $\pm$ 3.30%                 | 58.11 $\pm$ 2.30%                 | 53.00 $\pm$ 2.29%                 | 53.98 $\pm$ 0.81%                 |
|         | Contrastive | 57.81 $\pm$ 2.55%                 | 61.07 $\pm$ 7.20%                 | 55.64 $\pm$ 2.15%                 | 59.17 $\pm$ 7.90%                 | 57.73 $\pm$ 3.21%                 | 52.76 $\pm$ 1.75%                 | 56.81 $\pm$ 3.20%                 |
|         | K-Means     | 50.25 $\pm$ 0.23%                 | 50.46 $\pm$ 0.30%                 | 50.31 $\pm$ 0.16%                 | 50.33 $\pm$ 0.12%                 | 58.56 $\pm$ 2.65%                 | 55.41 $\pm$ 1.56%                 | 50.95 $\pm$ 0.18%                 |
|         |             |                                   |                                   |                                   |                                   |                                   |                                   |                                   |
| 0.6     | URE         | 88.77 $\pm$ 0.67%                 | 92.00 $\pm$ 0.98%                 | <b>78.81<math>\pm</math>0.66%</b> | 91.79 $\pm$ 2.61%                 | 59.75 $\pm$ 1.59%                 | 68.52 $\pm$ 0.62%                 | 64.43 $\pm$ 1.35%                 |
|         | ReLU        | 91.41 $\pm$ 0.16%                 | 93.73 $\pm$ 0.25%                 | 77.87 $\pm$ 0.88%                 | 95.33 $\pm$ 0.17%                 | 71.27 $\pm$ 1.11%                 | <b>76.03<math>\pm</math>0.62%</b> | 74.34 $\pm$ 0.95%                 |
|         | ABS         | <b>91.96<math>\pm</math>0.09%</b> | <b>94.03<math>\pm</math>0.21%</b> | 76.65 $\pm$ 1.35%                 | <b>96.34<math>\pm</math>0.18%</b> | <b>71.34<math>\pm</math>0.56%</b> | 75.61 $\pm$ 0.56%                 | <b>78.16<math>\pm</math>0.43%</b> |
|         | WCL         | 91.46 $\pm$ 0.25%                 | 93.55 $\pm$ 0.47%                 | 74.90 $\pm$ 0.37%                 | 94.65 $\pm$ 0.48%                 | 70.69 $\pm$ 0.67%                 | 71.00 $\pm$ 1.49%                 | 74.23 $\pm$ 2.04%                 |
|         | Siamese     | 60.09 $\pm$ 1.76%                 | 52.23 $\pm$ 0.29%                 | 67.89 $\pm$ 0.89%                 | 60.66 $\pm$ 1.94%                 | 59.19 $\pm$ 1.43%                 | 46.21 $\pm$ 5.86%                 | 57.51 $\pm$ 5.25%                 |
|         | Contrastive | 60.23 $\pm$ 0.60%                 | 63.69 $\pm$ 1.18%                 | 58.72 $\pm$ 1.08%                 | 60.17 $\pm$ 5.49%                 | 57.24 $\pm$ 0.71%                 | 52.73 $\pm$ 1.13%                 | 55.09 $\pm$ 1.94%                 |
|         | K-Means     | 50.58 $\pm$ 0.07%                 | 50.56 $\pm$ 0.50%                 | 50.66 $\pm$ 0.19%                 | 50.26 $\pm$ 0.18%                 | 59.10 $\pm$ 3.10%                 | 53.86 $\pm$ 0.45%                 | 50.96 $\pm$ 0.29%                 |
|         |             |                                   |                                   |                                   |                                   |                                   |                                   |                                   |
| 0.8     | URE         | 84.48 $\pm$ 0.64%                 | 91.07 $\pm$ 1.82%                 | 77.20 $\pm$ 3.21%                 | 88.51 $\pm$ 2.62%                 | 54.70 $\pm$ 2.53%                 | 67.00 $\pm$ 1.16%                 | 56.88 $\pm$ 1.09%                 |
|         | ReLU        | 86.45 $\pm$ 1.33%                 | 94.60 $\pm$ 0.44%                 | <b>81.23<math>\pm</math>0.22%</b> | 93.84 $\pm$ 0.50%                 | <b>80.84<math>\pm</math>0.53%</b> | <b>69.68<math>\pm</math>2.00%</b> | 67.36 $\pm$ 0.96%                 |
|         | ABS         | <b>86.71<math>\pm</math>1.13%</b> | <b>94.83<math>\pm</math>0.20%</b> | 80.83 $\pm$ 0.25%                 | <b>95.96<math>\pm</math>0.65%</b> | 79.94 $\pm$ 0.72%                 | 67.39 $\pm$ 1.85%                 | <b>68.79<math>\pm</math>4.73%</b> |
|         | WCL         | 86.52 $\pm$ 0.45%                 | 94.29 $\pm$ 0.32%                 | 73.90 $\pm$ 1.22%                 | 93.82 $\pm$ 0.22%                 | 69.45 $\pm$ 0.58%                 | 67.07 $\pm$ 0.99%                 | 68.53 $\pm$ 1.38%                 |
|         | Siamese     | 63.47 $\pm$ 4.52%                 | 57.80 $\pm$ 1.18%                 | 66.96 $\pm$ 1.09%                 | 53.20 $\pm$ 0.57%                 | 57.40 $\pm$ 2.42%                 | 53.77 $\pm$ 3.36%                 | 58.66 $\pm$ 3.62%                 |
|         | Contrastive | 71.20 $\pm$ 10.23%                | 82.10 $\pm$ 1.66%                 | 62.00 $\pm$ 10.86%                | 57.75 $\pm$ 0.12%                 | 55.51 $\pm$ 3.25%                 | 51.34 $\pm$ 0.69%                 | 56.11 $\pm$ 1.54%                 |
|         | K-Means     | 50.20 $\pm$ 0.09%                 | 50.59 $\pm$ 0.25%                 | 50.70 $\pm$ 0.19%                 | 50.28 $\pm$ 0.17%                 | 56.34 $\pm$ 3.78%                 | 55.18 $\pm$ 1.04%                 | 50.75 $\pm$ 0.43%                 |
|         |             |                                   |                                   |                                   |                                   |                                   |                                   |                                   |

TABLE II  
CLASSIFICATION ACCURACY (MEAN $\pm$ STD) OF THREE MOVA LEARNING METHODS ( $M = 4$ ) IS REPORTED FOR FIVE DATASETS, AVERAGED OVER 100 EPOCHS AND THREE TRIALS, UNDER THREE VARYING CLASS-PRIORS

| $\pi_p$ | Methods     | MNIST                             | Fashion-MNIST                     | Kuzushiji-MNIST                   | EMNIST-Digits                     | CIFAR-10                          | CINIC-10                          | SVHN                              |
|---------|-------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
| 0.4     | URE         | 91.57 $\pm$ 0.67%                 | 92.88 $\pm$ 0.25%                 | 78.52 $\pm$ 1.07%                 | 93.35 $\pm$ 0.34%                 | 64.85 $\pm$ 0.42%                 | 71.23 $\pm$ 1.15%                 | 58.23 $\pm$ 0.83%                 |
|         | ReLU        | 93.08 $\pm$ 0.26%                 | 94.67 $\pm$ 0.19%                 | 78.78 $\pm$ 0.69%                 | 94.84 $\pm$ 0.21%                 | <b>71.02<math>\pm</math>0.94%</b> | 74.81 $\pm$ 1.13%                 | 65.87 $\pm$ 3.20%                 |
|         | ABS         | <b>93.76<math>\pm</math>0.73%</b> | <b>95.56<math>\pm</math>0.27%</b> | <b>79.14<math>\pm</math>0.54%</b> | <b>95.94<math>\pm</math>0.08%</b> | 70.48 $\pm$ 0.61%                 | <b>76.05<math>\pm</math>1.71%</b> | <b>66.64<math>\pm</math>3.83%</b> |
|         | WCL         | 89.10 $\pm$ 1.03%                 | 91.32 $\pm$ 0.75%                 | 71.90 $\pm$ 0.23%                 | 91.83 $\pm$ 1.39%                 | 63.50 $\pm$ 1.07%                 | 62.86 $\pm$ 2.45%                 | 60.70 $\pm$ 0.86%                 |
|         | Siamese     | 56.09 $\pm$ 1.52%                 | 51.41 $\pm$ 0.36%                 | 54.27 $\pm$ 4.20%                 | 65.15 $\pm$ 0.23%                 | 51.35 $\pm$ 1.88%                 | 54.81 $\pm$ 13.15%                | 56.78 $\pm$ 3.01%                 |
|         | Contrastive | 54.32 $\pm$ 0.90%                 | 58.27 $\pm$ 1.25%                 | 56.69 $\pm$ 2.30%                 | 61.73 $\pm$ 7.92%                 | 55.28 $\pm$ 6.39%                 | 53.73 $\pm$ 3.51%                 | 53.08 $\pm$ 3.73%                 |
|         | K-Means     | 50.51 $\pm$ 0.16%                 | 50.20 $\pm$ 0.07%                 | 50.30 $\pm$ 0.23%                 | 50.19 $\pm$ 0.11%                 | 51.10 $\pm$ 0.27%                 | 50.63 $\pm$ 0.12%                 | 51.05 $\pm$ 0.76%                 |
|         |             |                                   |                                   |                                   |                                   |                                   |                                   |                                   |
| 0.6     | URE         | 89.78 $\pm$ 0.69%                 | 89.27 $\pm$ 0.23%                 | 75.74 $\pm$ 2.39%                 | 90.26 $\pm$ 1.91%                 | 56.32 $\pm$ 0.25%                 | 66.53 $\pm$ 0.28%                 | 56.83 $\pm$ 0.44%                 |
|         | ReLU        | 91.77 $\pm$ 0.61%                 | 92.50 $\pm$ 0.17%                 | 76.74 $\pm$ 0.31%                 | 92.85 $\pm$ 2.06%                 | 69.33 $\pm$ 0.26%                 | 74.81 $\pm$ 0.48%                 | 63.01 $\pm$ 0.17%                 |
|         | ABS         | <b>92.78<math>\pm</math>0.03%</b> | <b>93.61<math>\pm</math>0.20%</b> | <b>79.00<math>\pm</math>0.51%</b> | <b>95.46<math>\pm</math>0.20%</b> | <b>71.80<math>\pm</math>0.34%</b> | <b>75.61<math>\pm</math>0.56%</b> | <b>63.43<math>\pm</math>1.00%</b> |
|         | WCL         | 85.27 $\pm$ 0.91%                 | 90.65 $\pm$ 3.21%                 | 69.73 $\pm$ 1.71%                 | 90.78 $\pm$ 1.43%                 | 61.39 $\pm$ 1.83%                 | 57.62 $\pm$ 0.71%                 | 59.59 $\pm$ 0.45%                 |
|         | Siamese     | 62.17 $\pm$ 2.39%                 | 51.91 $\pm$ 0.52%                 | 59.11 $\pm$ 1.69%                 | 59.44 $\pm$ 0.76%                 | 56.28 $\pm$ 3.91%                 | 53.03 $\pm$ 4.57%                 | 55.25 $\pm$ 4.42%                 |
|         | Contrastive | 58.51 $\pm$ 0.53%                 | 62.90 $\pm$ 8.52%                 | 59.67 $\pm$ 2.31%                 | 59.06 $\pm$ 1.99%                 | 56.78 $\pm$ 2.17%                 | 51.81 $\pm$ 1.11%                 | 56.69 $\pm$ 2.84%                 |
|         | K-Means     | 50.40 $\pm$ 0.13%                 | 50.47 $\pm$ 0.11%                 | 50.42 $\pm$ 0.42%                 | 50.17 $\pm$ 0.08%                 | 51.21 $\pm$ 0.32%                 | 50.22 $\pm$ 0.19%                 | 50.72 $\pm$ 0.29%                 |
|         |             |                                   |                                   |                                   |                                   |                                   |                                   |                                   |
| 0.8     | URE         | 89.51 $\pm$ 2.59%                 | 88.77 $\pm$ 1.33%                 | 74.96 $\pm$ 1.04%                 | 88.20 $\pm$ 1.92%                 | 49.77 $\pm$ 0.80%                 | 67.44 $\pm$ 2.53%                 | 59.58 $\pm$ 3.15%                 |
|         | ReLU        | 93.73 $\pm$ 1.47%                 | 94.51 $\pm$ 0.69%                 | <b>81.20<math>\pm</math>0.34%</b> | 94.87 $\pm$ 0.04%                 | <b>79.45<math>\pm</math>0.50%</b> | 81.53 $\pm$ 1.30%                 | 78.86 $\pm$ 0.28%                 |
|         | ABS         | <b>93.88<math>\pm</math>0.42%</b> | <b>95.21<math>\pm</math>0.24%</b> | 80.96 $\pm$ 0.22%                 | <b>96.17<math>\pm</math>0.28%</b> | 78.83 $\pm$ 0.68%                 | <b>83.56<math>\pm</math>0.48%</b> | <b>79.13<math>\pm</math>0.22%</b> |
|         | WCL         | 80.75 $\pm$ 2.42%                 | 90.10 $\pm$ 1.01%                 | 65.18 $\pm$ 1.68%                 | 87.87 $\pm$ 0.21%                 | 58.96 $\pm$ 2.93%                 | 55.34 $\pm$ 1.04%                 | 58.20 $\pm$ 1.42%                 |
|         | Siamese     | 63.29 $\pm$ 2.63%                 | 54.31 $\pm$ 0.81%                 | 60.88 $\pm$ 4.32%                 | 53.99 $\pm$ 1.01%                 | 53.89 $\pm$ 4.64%                 | 54.20 $\pm$ 6.99%                 | 58.10 $\pm$ 3.42%                 |
|         | Contrastive | 68.39 $\pm$ 7.13%                 | 81.45 $\pm$ 1.37%                 | 68.24 $\pm$ 6.91%                 | 55.96 $\pm$ 1.79%                 | 58.14 $\pm$ 4.60%                 | 52.75 $\pm$ 0.94%                 | 53.41 $\pm$ 1.26%                 |
|         | K-Means     | 50.83 $\pm$ 0.87%                 | 50.72 $\pm$ 0.49%                 | 50.68 $\pm$ 0.55%                 | 50.21 $\pm$ 0.13%                 | 51.22 $\pm$ 0.90%                 | 50.56 $\pm$ 0.29%                 | 50.96 $\pm$ 0.19%                 |
|         |             |                                   |                                   |                                   |                                   |                                   |                                   |                                   |

consistently outperform the URE across all datasets and class prior settings. This superiority can be attributed to the effective risk correction mechanisms implemented in these methods, which help mitigate overfitting, particularly under conditions with imbalanced class priors.

The results further demonstrate that ReLU and ABS methods exhibit robust performance even as the class prior ( $\pi_p$ ) varies, maintaining higher classification accuracy compared to URE. This highlights the significance of the risk correction methods in improving the generalization ability of the model.

Notably, the performance of all methods shows some variation across different datasets, which is expected given the inherent differences in dataset characteristics. For instance, datasets like CIFAR-10, CINIC-10, and SVHN, which involve visually complex and diverse categories, showcase the ability of the ReLU and ABS methods to handle intricate multi-instance relationships. This capability allows the proposed methods to achieve superior classification accuracy compared to simpler datasets such as MNIST. Moreover, the sensitivity to class priors is particularly evident in Figs. 2–5. As  $\pi_p$  increases, there is a general decline in accuracy for all methods, but ReLU and ABS methods demonstrate a higher degree of resilience, sustaining competitive performance even under challenging prior distributions.

A critical observation from the loss curves presented in Figs. 2–5 is that URE often suffers from overfitting, with training loss values dipping into negative ranges. In contrast, both ReLU and ABS methods exhibit non-negative training losses throughout the training process, effectively regularizing the optimization process. This ability to maintain stable loss values contributes significantly to mitigating overfitting, ensuring that the models converge to better solutions. The generalization capabilities of the ReLU and ABS methods are further confirmed by their consistent performance across seven benchmark datasets, suggesting that these methods can adapt well to diverse data distributions.

The ReLU correction function excels in high-dimensional settings (e.g., CIFAR-10 with feature dimension of  $32 \times 32 \times 3$ ) by suppressing noise-induced negative risk, reducing computational overhead, and improving classification accuracy. However, its truncation mechanism discards negative risk gradients, which may degrade performance on small-scale or low-dimensional datasets (e.g., MNIST with feature dimension of  $28 \times 28$ ). In contrast, the ABS correction function preserves all risk information via absolute value operations, enabling robust decision boundaries and stable gradient updates. This makes ABS superior for low-dimensional tasks (e.g., handwritten digit classification), where negative risk gradients contain critical boundary cues. Experimental results (Tables I and II, Figs. 2–5) validate these trends: ReLU achieves higher accuracy on high-dimensional datasets, while ABS shows better stability and accuracy on low-dimensional datasets.

When compared to baseline methods, such as Siamese, contrastive, and K-means, the proposed methods provide substantial improvements in classification accuracy, particularly on CIFAR-10, where the ReLU method outperforms the best baseline by over 20%. This demonstrates that the proposed framework is not only effective but also significantly better than the traditional methods that do not leverage one-vs-all confidence comparison information among  $M$ -tuple data. These findings suggest that the proposed method is highly promising for practical applications, particularly in medical image analysis, where fully labeled data is often scarce and weak supervision represents the primary feasible option.

The stability of the proposed methods is further validated by the small standard deviations reported in Tables I and II, indicating that the methods perform consistently across different trials. This robustness, in combination with the

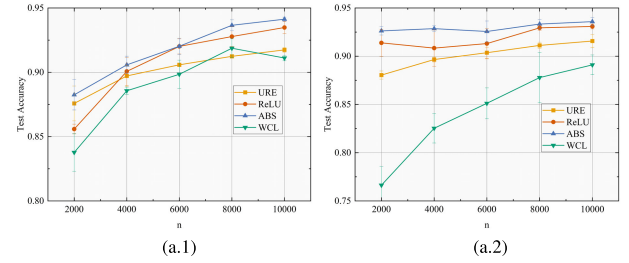


Fig. 6. Comparison of *Triple-MOVA* and *Quadruple-MOVA Learning* performance under different training sample sizes ( $\pi_p = 0.4$ ). (a.1) (Triple) MNIST. (a.2) (Quadruple) MNIST.

superior performance observed, demonstrates the reliability and applicability of the ReLU and ABS methods in diverse weakly supervised learning scenarios. Such consistency is particularly crucial when applying these methods in real-world tasks, where data and model behavior can be unpredictable.

On low-dimensional datasets, the ABS correction method is more advantageous as it retains negative risk information that aids in capturing fine decision boundaries; while on high-dimensional datasets, the ReLU correction method is relatively superior due to its truncation mechanism that suppresses overfitting from redundant features and noise. However, this trend is not absolute, since larger tuple sizes allow negative risk to carry richer confidence comparison information, enabling the ABS method to outperform ReLU on some high-dimensional datasets under certain class-prior conditions. Therefore, the performance differences between the two correction approaches are influenced not only by dimensionality but also by dataset characteristics, tuple size, and class-prior settings.

Fig. 6 presents a comparison between *WCL* and *MOVA learning* (*Triple-MOVA* and *Quadruple-MOVA*) on the MNIST dataset under the setting  $\pi_p = 0.4$  across different training sample sizes. The results clearly show that *MOVA learning* significantly outperforms *WCL* under small-sample and higher order tuple conditions, with the performance gap becoming more pronounced for quadruples. As the number of training samples increases, *WCL* gradually catches up under the triple-tuple setting, but remains consistently inferior under the quadruple setting. Among our proposed methods, both ReLU and ABS demonstrate superior performance. Moreover, Fig. 6 shows that *WCL* exhibits larger fluctuations and less stable convergence under small-sample regimes, further illustrating its difficulty in handling weak supervision. While both *WCL* and our methods improve with more training data, *WCL* consistently lags behind.

In addition, we systematically compare *WCL* with the three proposed *MOVA* variants (URE, ReLU, and ABS) across four types of label noise: symmetric, asymmetric, pure tuple, and mixed noise. The results (Figs. 7 and 8) show that *MOVA* methods outperform *WCL* under small-sample and low-to-moderate noise settings, particularly in higher order tuples. As noise intensity increases, however, *WCL* exhibits stronger stability, benefiting from the consistency of positive pairs, and in several high-noise scenarios even surpasses *MOVA*. These results highlight complementary strengths: *MOVA* excels at

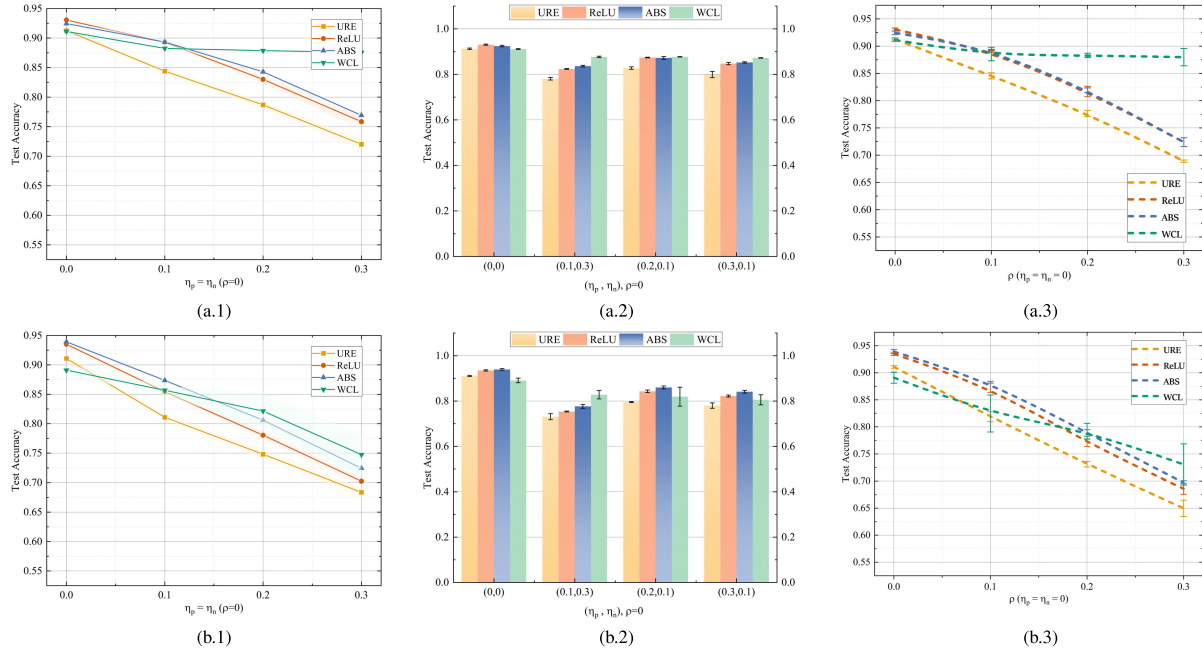


Fig. 7. Comparison of *Triple-MOVA* and *Quadruple-MOVA* Learning on the MNIST dataset under symmetric, asymmetric, and pure tuple noise conditions ( $\pi_p = 0.4$ ). (a.1) (Triple) MNIST-symmetric noise. (a.2) (Triple) MNIST-asymmetric noise. (a.3) (Triple) MNIST-pure tuple noise. (b.1) (Quadruple) MNIST-symmetric noise. (b.2) (Quadruple) MNIST-asymmetric noise. (b.3) (Quadruple) MNIST-pure tuple noise.

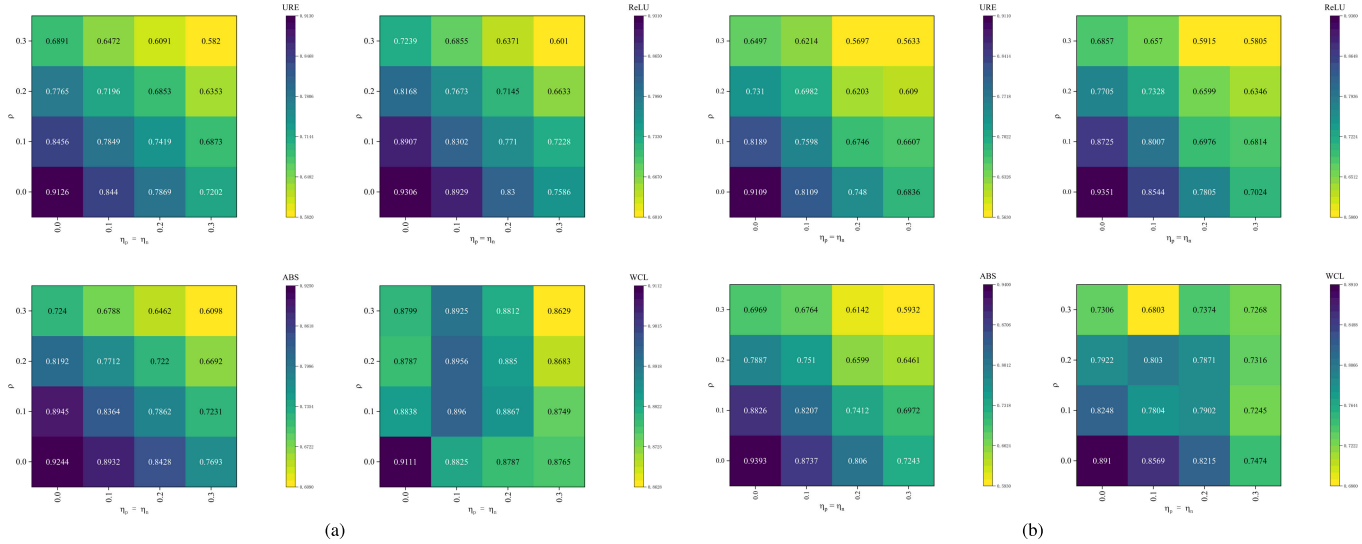


Fig. 8. Classification performance of *Triple-MOVA* and *Quadruple-MOVA* on the MNIST dataset under mixed noise conditions ( $\pi_p = 0.4$ ). (a) (Triple) MNIST-mixed noise ( $\pi_p = 0.4$ ). (b) (Quadruple) MNIST-mixed noise ( $\pi_p = 0.4$ ).

fine-grained supervision when noise is limited, while *WCL* is more robust under severe noise.

In summary, the risk estimation paradigm offers a more direct and theoretically guaranteed solution in weakly supervised settings, making it particularly suitable when computational resources are limited. On the other hand, the representation learning paradigm has advantages in large-scale, multitask, and diverse-distribution scenarios. Our experimental results are consistent with these theoretical insights: under weak supervision and small-sample conditions, the proposed risk estimation methods outperform contrastive learning methods.

## VI. CONCLUSION

This study proposes a generalized weakly supervised learning framework termed *MOVA learning*. Leveraging comparative relationships among multiple samples, this work establishes a URE and derives a generalization error bound that guarantees optimal parameter convergence rates. To mitigate overfitting issues inherent in UREs for classification tasks, two correction functions are introduced to derive corrected risk estimators. Empirical experiments on seven benchmark datasets demonstrate that the corrected risk estimators consistently outperform their unbiased counterparts in classification performance.

## VII. FUTURE WORK

Future research will extend the proposed *MOVA learning* framework along two directions. First, we aim to generalize the binary  $M$ -tuple confidence comparison to multiclass scenarios. Similar to the binary case, the generation of multiclass  $M$ -tuples imposes no explicit constraints on class attributes; it only requires that the dominant sample exhibits higher confidence in some class  $k$  than all others, i.e.,  $p(y^1 = k|x^1) > \max_{j=2,\dots,M} p(y^j = k|x^j)$ . The resulting URE holds symmetrically across all classes, ensuring generalization. This unconstrained generation rule avoids introducing extra prior knowledge during data construction and maximizes consistency with real weak supervision, thereby demonstrating the scalability of *MOVA* to multiclass tasks.

Second, while the current *MOVA* framework is designed for unconstrained, small-to-medium-scale risk minimization problems where first-order optimizers such as Adam suffice, larger scale or constrained scenarios call for more advanced techniques. The distributed inertial proximal neurodynamic method [49] can accelerate convergence and reduce memory costs in large-scale  $M$ -tuple training, and the accelerated primal-dual projection neurodynamic method [50] shows strong performance in convex or sparsity-constrained optimization. Since the objectives studied here are unconstrained, the primal-dual strategy is not directly applicable and cannot mitigate overfitting caused by negative risk terms, which are instead addressed by our risk-correction functions. Nevertheless, both approaches remain promising for extending *MOVA* to distributed, large-scale, or constrained optimization tasks in future work.

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